Laser Joint

Created by:

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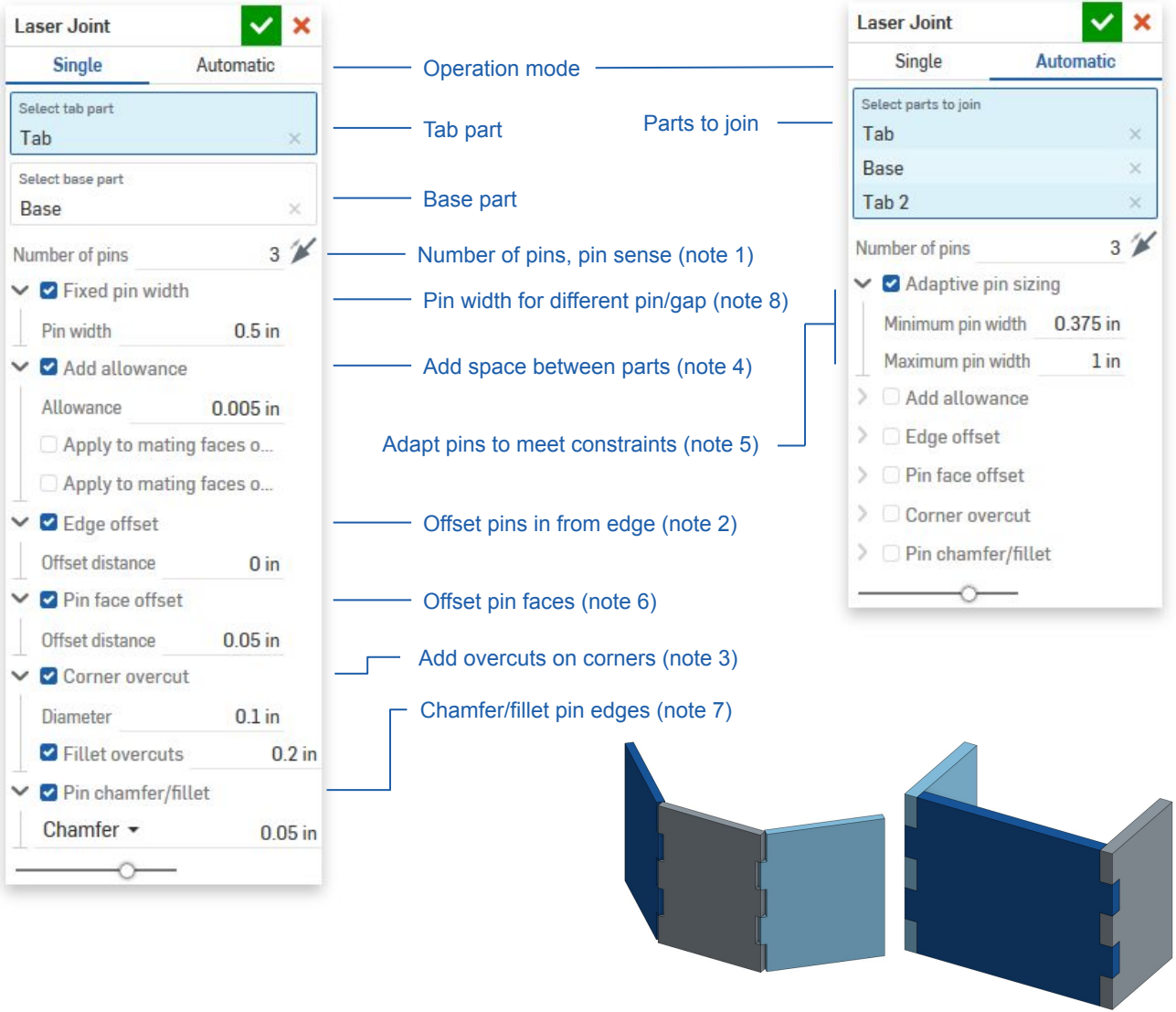
Description:

This custom feature creates a laser-cuttable finger joint between intersecting planar parts.

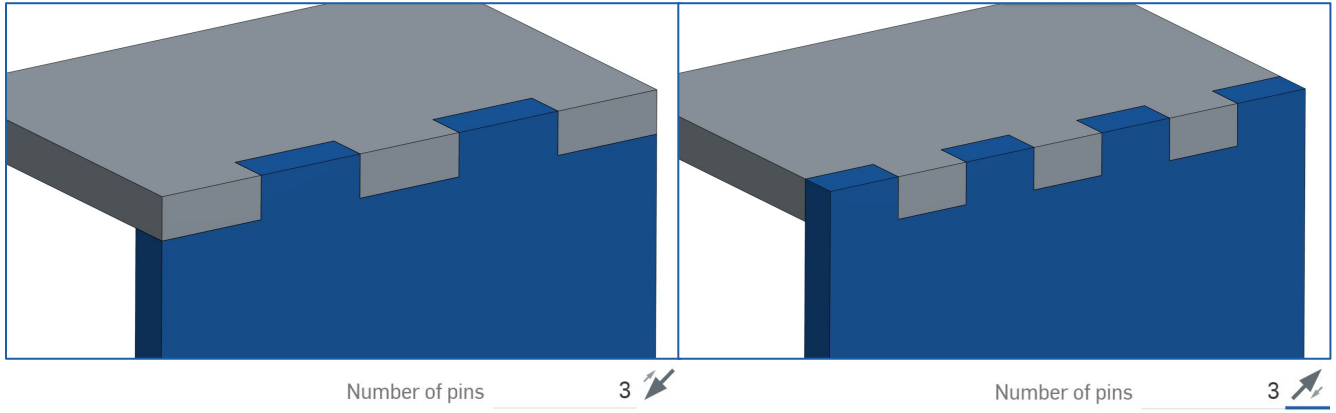
Implements manual and automatic joint generation, with options for corner overcuts, allowances for fit tolerancing, and adaptive sizing of pins.

Prerequisites:

Two or more intersecting planar parts that can be cut by laser. Feature assumes the largest face defines the laser cutting plane.



Note 1: Pin Sense

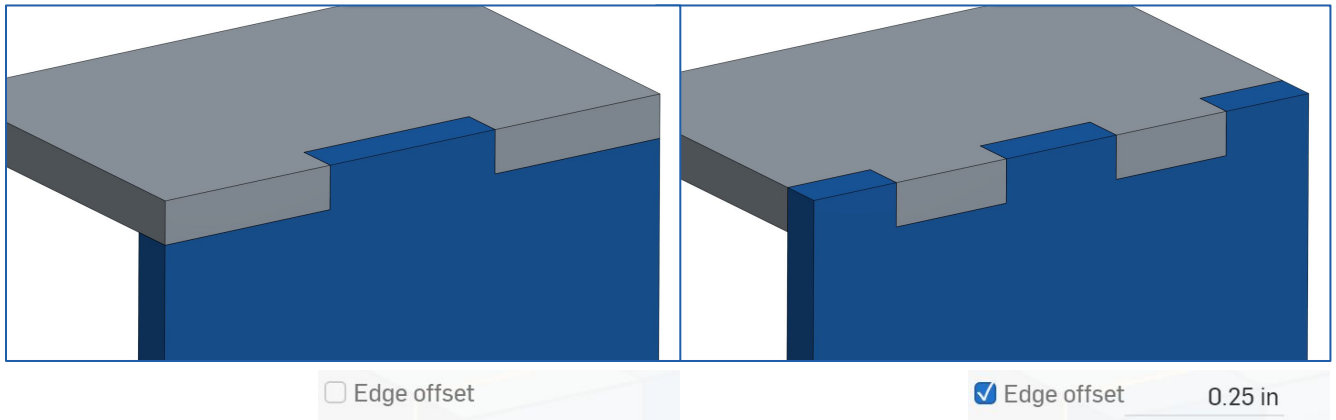


Toggling pin sense toggles whether the tab part has pins or slots at the edges.

In the images at left, the gray tab mates with the blue base part.

Edge offsets are applied before determining the position of the first pin.

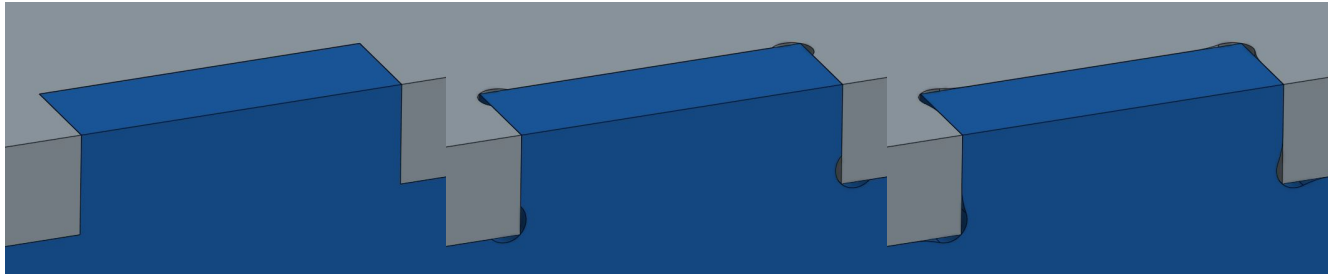
Note 2: Edge Offset



Applying an edge offset symmetrically moves the pins in from either end of the joint.

Here the gray tab has two pins in either case, but with an edge offset in the right image.

Note 3: Corner Overcut



☐ Corner overcut	☑ Corner overcut	☑ Corner overcut
	Diameter 0.1 in	Diameter 0.1 in
	☐ Fillet overcuts	☑ Fillet overcuts 0.2 in

Applying a corner overcut cuts a cylindrical clearance at each concave corner created by the joint in both tab and base parts.

The sharp corners resulting from these overcuts may have an optional fillet applied.

Note 4: Allowance

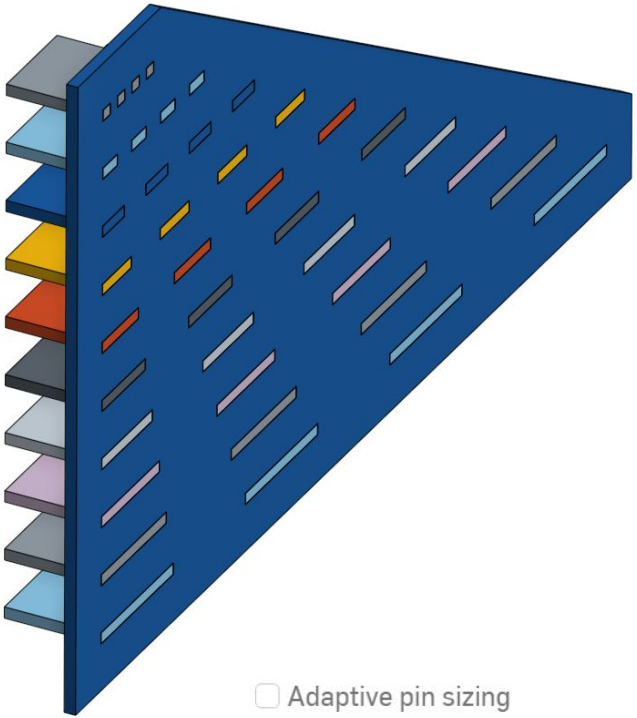
☐ Add allowance	☑ Add allowance	☑ Add allowance	☑ Add allowance	☑ Add allowance
	Allowance 0.01 in	Allowance 0.01 in	Allowance 0.01 in	Allowance 0.01 in
	☐ Apply to mating faces on tab	☑ Apply to mating faces on tab	☐ Apply to mating faces on tab	☑ Apply to mating faces on tab
	☐ Apply to mating faces on base	☐ Apply to mating faces on base	☑ Apply to mating faces on base	☑ Apply to mating faces on base

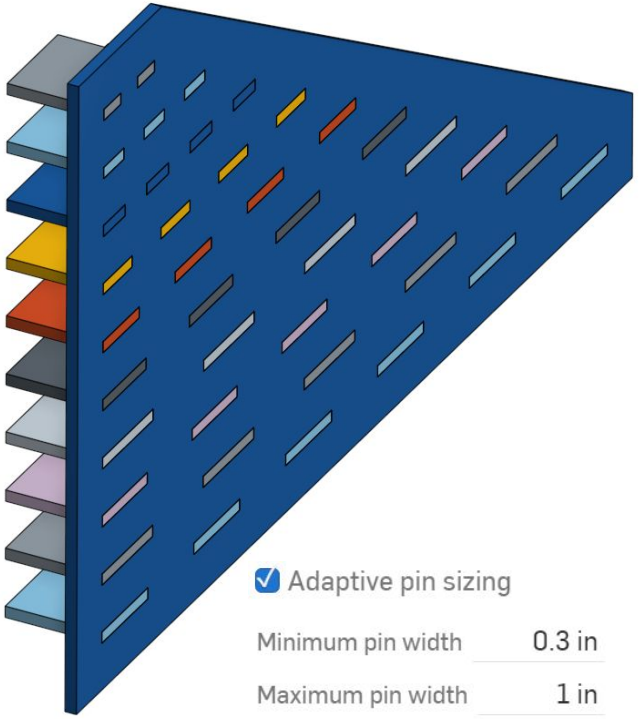
Adds an allowance to meet tolerances for the fit of the joint.

Allowance amount is applied to the faces internal to the joint, and optionally to one or both of the mating faces of the tab and base parts.

In the images at left, the gray part is the tab, blue is the base.

Note 5: Adaptive Pins



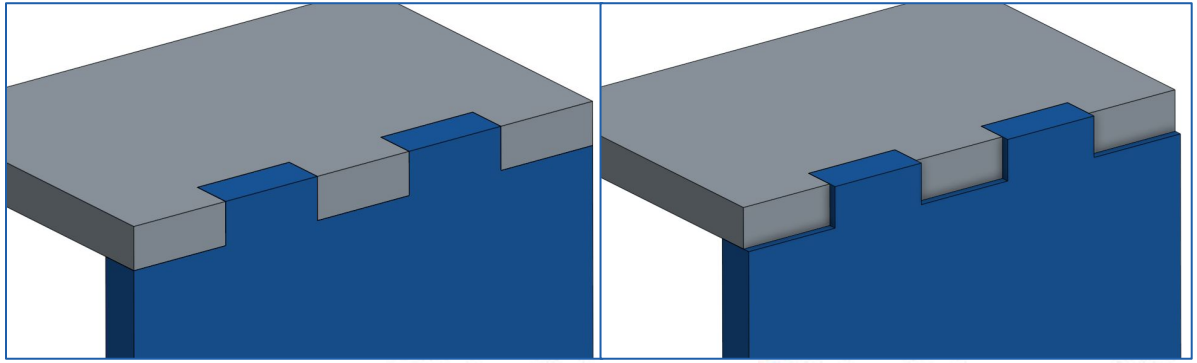


Enabling adaptive pins computes the number of pins required to maintain pin width within the given bounds. This is only available in automatic joint generation, where it is useful for joining parts ranging widely in scale.

In the image at left, without adaptive sizing the smallest tab has very fine pins while the largest tab still has too few. Enabling adaptive pins at right results in pins that are within the chosen size constraints.

Note that enabling this option overrides the “Number of pins” input field, though it does not override pin sense.

Note 6: Offset Pin Faces



☐ Pin face offset

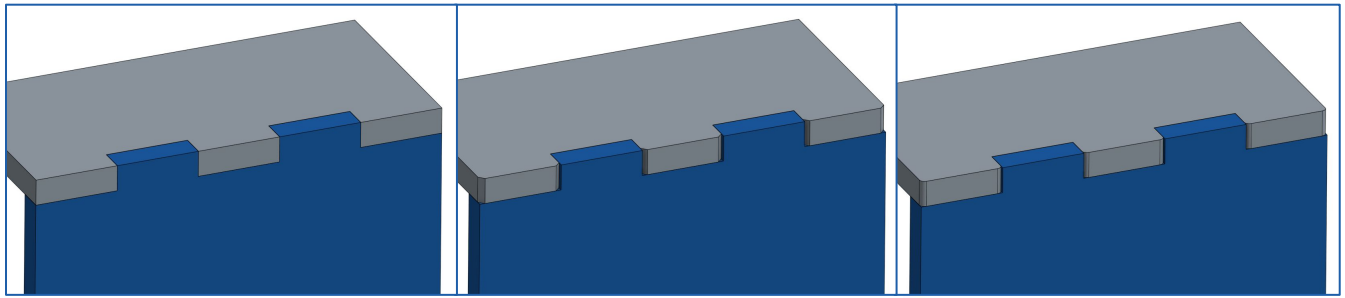
☑ Pin face offset 0.05 in

Toggling pin face offset toggles whether the end faces of the tab pins are offset. Values may be positive or negative.

In the images at left, the gray tab mates with the blue base part.

Note that only pin faces wholly contained within the joint will be offset.

Note 7: Pin Chamfer/Fillet



☐ Pin chamfer/fillet

☑ Pin chamfer/fillet

Chamfer ▾ 0.05 in

☑ Pin chamfer/fillet

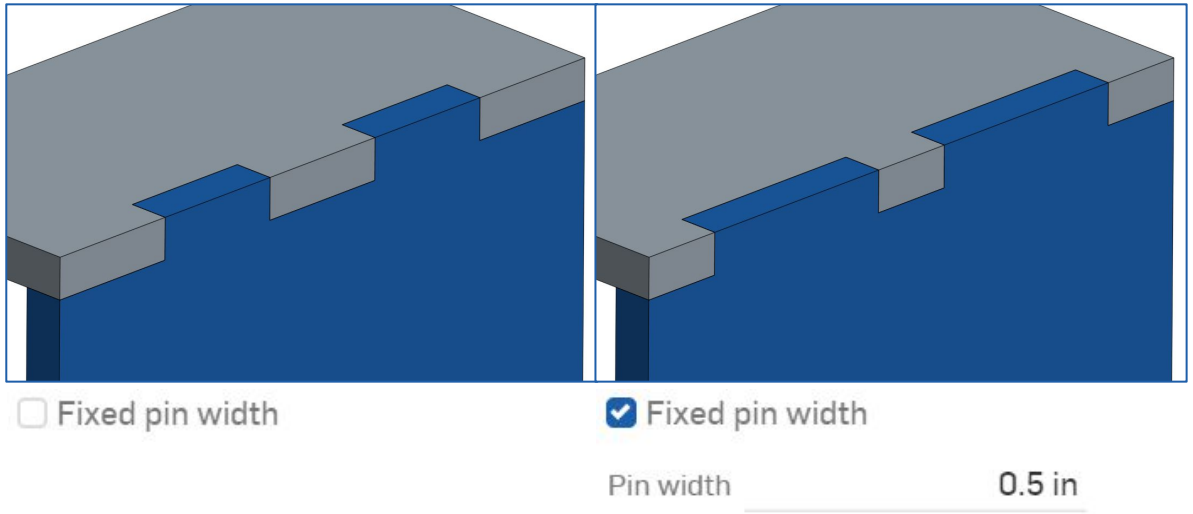
Fillet ▾ 0.05 in

Toggling this option enables or disables chamfer/fillets on the insertion edges of the tabs.

In the images at left, the gray tab mates with the blue base part.

Chamfer/fillet is applied to convex edges of the pins created by the joint.

Note 8: Fixed Pin Width



Enabling defined pin width is available when in single joint mode of operation. When enabled, this allows the user to define a specific number of pins and a specific width of each pin.

In the images at left, the gray tab mates with the blue base part.

Will throw an error if the requested number of pins are longer than the length of the joint.